When Simple Is Enough: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction

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Abstract

Small bioactivity data sets make it easy to overstate the benefit of complex models, through data leakage and through model selection that is rarely evaluated on unseen data. We re-examined both for inhibitors of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR, $\boldsymbol{N = 859}$ curated compounds). Removing the information leaks of the original study's code one at a time lowered its $\boldsymbol{R^2}$ from 0.853 to 0.740, the level of a model with no feature selection; the largest contribution was a feature importance computed on the test molecules. The same leak on a classical learner reproduced the reported gain (0.836 against 0.748 when kept clean).

Leakage-free, four classical representations and seven neural models were scored on the same 25 outer folds of a repeated nested cross-validation. Morgan-fingerprint ensembles reached $\boldsymbol{R^2 = 0.769 \pm 0.041}$, and ten 3D shape descriptors added to the best descriptor set changed nothing ($\boldsymbol{\Delta R^2 = 0.001}$). A Directed Message Passing Neural Network reached 0.725 by default and 0.726 after a nested search, below the fingerprint ensembles by a margin that cleared multiple-comparison correction only narrowly (smallest adjusted $\boldsymbol{p = 0.041}$). Only a network initialized from a pretrained foundation model matched them (0.761,

equivalent in both metrics), at eight times the cost; none exceeded them. Scaffold-grouped splits cost every model between 0.16 and 0.24 units of R^2. On an FDA-approved library the deployed model placed only 14 of 3056 scored compounds inside its applicability domain (median similarity 0.17 against 0.70 internally), so the compounds it ranks highest are hypotheses rather than leads. For this target a fixed-fingerprint ensemble was sufficient; we make no claim beyond it.

Keywords: Quantitative Structure-Activity Relationship, Low-Data Regimes, Nested Cross-Validation, Data Leakage, Graph Neural Networks, Applicability Domain, Model Selection Bias

1 Introduction

1.1 Predictive Modeling in Low-Data Bioactivity Regimes

Quantitative structure–activity relationship (QSAR) modeling is constrained when few labeled compounds are available [1]. In that regime, simpler and more interpretable approaches, such as multivariate linear regression or classical tree-based ensembles, often match or exceed the predictive reliability of neural networks [2, 3]. These models carry more bias and less variance, and they are less sensitive to the random composition of the training sample than overparameterized alternatives, particularly when they are given descriptors derived from quantum-chemical calculations [3].

Small data sets invite overfitting, and the risk grows as the number of molecular descriptors approaches or exceeds the number of compounds [4, 5]. One common response is to enlarge the training set by data augmentation or synthetic sample generation [6, 7], which costs computation [7, 8] and, applied carelessly, obscures what the base algorithm can do on its own. When augmentation precedes partitioning, the "augment-then-split" fallacy, perturbed copies of a test molecule reach the training set; the partitions are then no longer independent and the reported metrics are inflated [6, 9, 10]. Augmentation therefore has to stay inside the training folds of a partition fixed beforehand [6, 9].

Evaluating complex architectures in this regime also requires separating model building from model assessment. Tuning hyperparameters on a small sample with a large search space biases the selection [11], so an architecture scored on the same data that chose it says little about its behavior on unseen compounds. Double, or nested, cross-validation is the established way to decouple the two [11, 12]. Even then, a random k-fold partition measures interpolation among related compounds rather than extrapolation to new chemotypes [13, 14], and generative deep learning extrapolates poorly in the sparse, discrete space of Simplified Molecular-Input Line-Entry System (SMILES) strings [15].

A model fitted to few compounds can also reach a good score by chance correlation, which an empirical check can detect. In y-randomization the response is permuted and the model refitted, so that a score close to the one obtained with the true responses points to chance correlation rather than to a learned relationship [16]. Rücker et al.

prescribe more than permuting the response: the whole model-building procedure, descriptor selection included, should be repeated inside each randomization, and the result compared with models built on random descriptors [16]. Where curated descriptors pass these checks, machine learning has been used to rank candidate compounds, for instance to separate amyloid-β aggregation inhibitors from decoys [17].

Transfer learning offers a different route around data scarcity, by adapting networks trained on a large source domain [18]. Reduced learning rates during fine-tuning [19] and lighter convolutional architectures in place of Transformers [20] have been reported to help with class imbalance and selection bias in low-resource settings.

1.2 Domain-Engineered Features versus End-to-End Representation Learning

Feature-based machine learning relies on domain-engineered descriptors and molecular fingerprints [21]. Morgan (ECFP4) fingerprints encode local substructures, while physicochemical and 3D descriptors quantify specific molecular properties; both are computed deterministically and need no large training corpus [3, 17, 22]. Their drawback is dimensionality: accumulating handcrafted features produces large and redundant matrices [2, 5], which on restricted data sets raises computational cost and makes feature selection itself a source of error [2, 5].

End-to-end representation learning avoids manual feature engineering. Variational autoencoders (VAEs), generative adversarial networks (GANs) and graph neural networks (GNNs) map molecular topologies into continuous latent spaces and model nonlinear interactions without a predefined descriptor set [15, 21, 23], and chemical language processing (CLP) applies the same idea to string representations such as SMILES [20]. The discrete nature of those strings is a weakness of the approach, since altering a single character can produce an invalid structure [15].

How much this flexibility is worth when labels are scarce remains unsettled. Representation learning needs enough examples to recover what a descriptor encodes by construction, which makes it less effective as the labeled set shrinks [24]. For inhibitors of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR), Karimova and Senanayake [25] addressed this limitation with Gaussian noise-based data augmentation and ensembles of deep neural networks, raising their baseline R^2 from 0.75 to 0.85, and released both the curated data set and the code that produced it. Their result poses the question this study takes up: on a set of this size, is synthetic data generation necessary for predictive reliability? Classical ensembles are reported to match or exceed deep architectures when data are scarce [2, 3, 18, 21, 22, 26], although such comparisons are rarely run on identical, leakage-free folds.

Opinion on nonlinear models in limited-data settings is itself shifting [27]. Regularized nonlinear algorithms combined with structured linear priors reach results comparable to classical methods [26, 27], and fine-tuning pretrained networks such as ChemBERTa has produced usable predictions from a few dozen labeled compounds [18, 28].

1.3 Data Leakage in Low-Data Molecular Modeling

Data leakage, the entry of test-set information into the training process, inflates performance estimates without producing an error or a warning [6, 29]. In low-data QSAR it arises most often in two places. The first is preprocessing: when scaling, imputation or variance-based filtering is fitted on the complete data set, the model is given information about the distribution of the test compounds, the independence of the partitions is lost, and the resulting estimate is optimistic [30, 31].

The second is model selection. A validation set that repeatedly guides hyperparameter choices stops being independent of the model, and highly parameterized architectures such as deep neural networks are the best placed to exploit that overlap, fitting noise and returning optimistic estimates of the generalization error [12, 24]. Both paths are closed by fitting every transformation and every selection step inside the training partition at hand. Because the details of the partitioning are often reported incompletely, performance is sometimes attributed to a new architecture when it comes from leakage instead [32].

1.4 Scope and Contributions

This study asks whether, on one small and well-characterized bioactivity set (859 TgDHFR inhibitors), graph networks, 3D descriptors, feature selection and data augmentation improve on classical ensembles trained on fingerprints once the evaluation is free of leakage. We do not assume the answer; the evaluation is designed so that either outcome would be detectable. Every model is scored on the same 5×5 repeated nested cross-validation folds, with each preprocessing and selection step fitted inside the training partitions, and every comparison is made with the Nadeau–Bengio corrected t-test under Holm adjustment and with an equivalence test against a pre-specified margin, so that "no significant difference" is distinguished from "demonstrated equivalence". The conclusions are checked under scaffold-grouped splits, with the extreme-potency records retained, and against four forms of y-randomization. On that basis we contribute:

- **A quantified taxonomy of leaks.** Four ways in which information about the test molecules reaches a model in low-data molecular machine learning—re-partitioning after augmentation, early stopping on the test set, feature importance computed in sample, and duplicate structures—are removed one at a time from a published pipeline, so that the contribution of each to the reported R^2 is measured rather than asserted.
- **The reported gain, reproduced as an artifact.** The feature selection and Gaussian-noise augmentation that took the original R^2 from 0.75 to 0.85 do not improve a classical learner under the same hold-out protocol, but the same selection performed leakily on that learner reproduces the improvement, so the gain measures the leak and not the intervention.
- **A leakage-free comparison with equivalence testing.** Four representation spaces are compared against seven neural models—a Chemprop D-MPNN with default and with searched hyperparameters, a seed ensemble, a descriptor-augmented network, a feed-forward network on the same fingerprints, a pretrained

foundation model, and the original architecture rebuilt without its leaks—on identical folds and with their measured computational cost. A pretrained model is the only network that reaches equivalence with the fingerprint ensemble.

The final model is also applied to a library of approved drugs, which locates the candidates it ranks highest with respect to its applicability domain. Those are computational hypotheses and are reported as such.

2 Materials and Methods

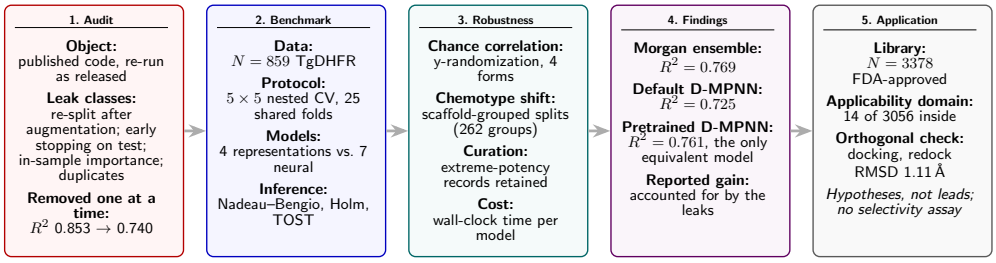


Fig. 1 Structure of the study. The published pipeline is first re-executed and taken apart one leak at a time, which measures what each class of leak contributes to its reported R^2 (1). The same data are then used for a leakage-free benchmark in which every model is scored on identical outer folds and every comparison is corrected for the overlap of training sets and tested for equivalence (2), and the conclusions are checked against chance correlation, a harder splitting scheme, the curation decision and computational cost (3). The findings (4) are that a fingerprint ensemble is not outperformed by a graph network at this data volume, that only a pretrained network reaches equivalence with it, and that the gain reported by the original study is accounted for by the leaks removed in (1). The final model is applied to approved drugs as hypothesis generation, with each candidate placed relative to the applicability domain (5).

An overview of the multi-tier virtual screening pipeline implemented in this study is illustrated in Figure 1.

2.1 Bioactivity Benchmark Dataset and Curation

This study utilizes a previously published benchmark of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) enzyme inhibitors [25]. Adopting the baseline dataset compiled by Karimova and Senanayake establishes a rigorous methodological control. This framework isolates the predictive capacity of classical machine learning algorithms from computationally heavy statistical interventions. Omitting synthetic data augmentation during the primary benchmarking phase ensures that performance metrics reflect empirical ground-truth data rather than algorithmic artifacts.

Initial records merged drug-protein interaction parameters and SMILES structures from BindingDB with bioactivity data from ChEMBL [25]. Automated standardization protocols, including salt stripping and structural validation, ensured chemical integrity [19]. The merged table contained 875 curated records. Extreme potency values were screened with Tukey's rule on their pIC_{50} distribution (fences at Q1 − 1.5 IQR

$= 2.06$ and Q3 + 1.5 IQR $= 10.80$). 2 records fell outside the fences: a classical antifolate analogue at $pIC_{50} = 11.68$ (high fence) and a practically inactive compound at $pIC_{50} = 1.56$ (low fence). They were excluded from the primary analysis, which leaves 873 entries, the size reported in the original study [25]. Because an exclusion of this kind can itself shape the reported performance, the full benchmark was also run with these two records retained (Section 3.5).

For compounds exhibiting multiple IC_{50} measurements across varying experimental conditions, we applied the Interquartile Range (IQR) method, as in the original study. Duplicate assays with a bioactivity variance exceeding 1.5 log units were discarded as statistical outliers [25]. The remaining consistent entries were averaged to yield a single representative measurement per ligand.

To adhere to the Organization for Economic Co-operation and Development (OECD) principles for QSAR validation [33–35] and to keep the same structure out of both training and test folds, all SMILES were then canonicalized with RDKit. The 873 entries turned out to describe only 859 distinct structures: 14 molecules appeared twice under different PubChem CIDs, and the second occurrence of each was removed (the first was retained). The measured pIC_{50} values within each pair differed by at most 0.18 units, so the target distribution is practically unchanged. The removed records are listed in `data/V2-combined_bd_chmbl_unique_duplicates.csv`.

This curation protocol yielded a final benchmark of 859 unique small molecules. The continuous target variable was transformed into pIC_{50} ($-\log_{10}(IC_{50})$) [25]. This logarithmic conversion standardizes response values and compresses the active range, optimizing the target distribution for robust regression modeling.

To explicitly quantify the performance impact of the interventions reported in the original literature, we replicated its hold-out protocol in an isolated ablation study. A 15% hold-out partition was separated first (seed 42), and everything below was learned from the development partition only. The baseline (A) is the stacking ensemble on all 2048 Morgan bits. The feature-selection variant (B) computes the permutation importance of every bit on the development partition (5 repeats, negative mean squared error) and keeps the bits whose mean importance is positive beyond floating-point noise (greater than 10^{-9}), because bits that the model never uses have an importance that is zero up to rounding error. The augmentation variant (C) adds, to the selected features, perturbation of the training features via additive Gaussian noise: applying the two noise levels of the published code ($\sigma = 0.01$ and $\sigma = 0.001$, which that code introduces as a factor to be adjusted to the scale of the data and does not otherwise justify) generated two synthetic variants per original molecule, which effectively triplicated the training space (to 2190 rows) while maintaining absolute isolation of the empirical test set. Each variant is the average of five stacking ensembles that differ in the seed of their tree models.

The leak that the original selection contains was also measured on this classical replication, where it needs no deep network. Variant B computes its permutation importance on the development partition alone. A second version of it, identical except that the importance is computed on the complete data set as the original study does, was fitted and scored on the same held-out molecules, and the two sets of selected bits were compared by their Jaccard index. Variant C was treated in the same way.

2.2 Audit of the Original Pipeline

The comparison above treats the figures of the original study as given. To establish what produces them, its published code was obtained and run in its original TensorFlow environment before anything was changed, which fixes the object of the audit as the pipeline as published rather than a reimplementation of it. Because one run of a stochastic pipeline carries its own variability, every stage was then repeated over 10 random seeds.

The pipeline was then taken apart one leak at a time. Starting from the published protocol (R0), four modifications were applied in sequence, each removing one path by which information about the test molecules reaches the fitted model and changing nothing else: (R1) the synthetic copies produced by Gaussian augmentation are no longer re-partitioned after augmentation, so a perturbed copy of a test molecule can no longer train the model; (R2) early stopping and model choice use an internal split of the training partition instead of the test set; (R3) the permutation importance that drives feature selection is computed out of fold inside the training rows instead of on the test molecules; and (R4) structures that occur more than once are removed from both sides of the split. Every rung was run over 10 seeds. Rungs R1 to R4 are scored on the 130 molecules common to all of them, so their differences are paired over seeds and over molecules and a percentile bootstrap over those molecules gives their confidence intervals; the published protocol builds its own evaluation partition and is therefore not paired with them. A reference model with no feature selection of any kind was fitted on the same clean split.

2.3 Representation Spaces: Molecular Fingerprints and Conformational Descriptors

We generated four distinct molecular representation spaces using RDKit [36] to systematically map the structure-activity relationship. The structural baseline relied on Morgan fingerprints (ECFP4). We computed these with a radius of 2 and folded them into a fixed-length vector of 2048 bits. Extended-connectivity fingerprints offer rapid calculation and capture a vast array of distinct local substructural features [37]; chirality was not encoded. Classical 2D fingerprint-based models frequently achieve predictive parity with state-of-the-art 3D structure-based models when relying exclusively on ligand information [38].

To establish a physicochemical baseline, the pipeline extracted the complete RDKit descriptor list (217 conventional two-dimensional topological and physicochemical descriptors). Curated 2D molecular representations demonstrate substantial robustness; recent studies confirm they maintain scaffold-independent advantages over 3D alternatives across various property prediction tasks [39]. A third hybrid space concatenated these 217 descriptors with the 2048-bit Morgan fingerprints to evaluate potential synergistic effects.

A fourth representation appended ten 3D shape descriptors (asphericity, eccentricity, inertial shape factor, the two normalized principal moment ratios, the three principal moments of inertia, radius of gyration, and spherocity index) to this hybrid, for 2275 features in total, to assess the predictive value of spatial geometry. Hydrogens

were added and a single conformer per molecule was generated with the Experimental-Torsion Knowledge Distance Geometry algorithm (ETKDGv3, fixed random seed of 42, random initial coordinates as a fallback) [40]. Because purely distance-based constraints often distort aromatic rings and sp^2 centers, ETKDGv3 integrates experimental torsion-angle preferences derived from small-molecule crystallographic data. The conformer was then locally minimized with the MMFF94 force field before descriptor extraction. Conformer embedding failed for 0 of the 859 molecules, and 0 descriptor values of the 2D+3D+FP matrix were missing or infinite before imputation.

Although integrating spatial information remains critical for molecular image-based modeling [41] and geometry-augmented representation learning [42], recent empirical evaluations suggest that 3D conformer descriptors yield limited incremental predictive value over curated 2D representations in standard tabular benchmarks [39]. Our nested cross-validation framework explicitly tested this hypothesis. The purely structural Morgan fingerprints yielded a mean R^2 of 0.769, without a significant difference from the 3D hybrid space ($R^2 = 0.774$; Nadeau-Bengio-corrected paired t-test on the per-fold R^2, $p = 0.4006$). The two best-scoring spaces, the 2D descriptors with fingerprints and the 3D hybrid, were likewise indistinguishable from each other ($p = 0.8243$). Since no space was reliably superior and the Morgan space needs neither descriptor calculation nor conformer generation, it was adopted for all downstream feature selection, data augmentation, and virtual screening protocols.

2.4 Nested Cross-Validation Protocol and Variance Correction

Model building and model assessment were separated by a 5×5 repeated nested cross-validation with random seed 42: a 5-fold outer loop repeated 5 times, giving 25 outer evaluations, with a 5-fold inner loop inside each outer training partition [11, 43]. Methodological frameworks do not require the two loops to use the same number of folds [43], and a systematic comparison of validation schemes finds no uniformly superior variant, its ranking depending on the algorithm, the data and the circumstances of the study [44]. The same outer partitions were used for every representation and every model, so that all comparisons reported here are paired. Figure 2 shows one outer fold of this protocol.

Random splitting was adopted for the main benchmark. The objective is interpolation within the applicability domain of known TgDHFR inhibitors rather than extrapolation to new chemotypes, and at this sample size ($N = 859$) scaffold-based partitioning produces imbalanced folds from which whole bioactivity ranges can disappear [13]. Because random splits are optimistic for series of close analogs [45, 46], the Morgan-fingerprint ensemble and the D-MPNN were also run under scaffold-grouped splits. The Bemis-Murcko scaffold of each compound, without stereochemistry, defined a group; the 859 compounds fall into 262 groups, the largest with 49 members and 159 containing a single compound. In each of the 5 repeats, groups were visited in random order and each was assigned to the currently smallest of the 5 folds, so that no scaffold appears on both sides of an outer split. Everything else was identical to the random-split protocol, and both model families again shared the same folds.

The inner loop selected the architecture and, at the same time, confined every data-dependent transformation to the partition it was fitted on. Within each outer

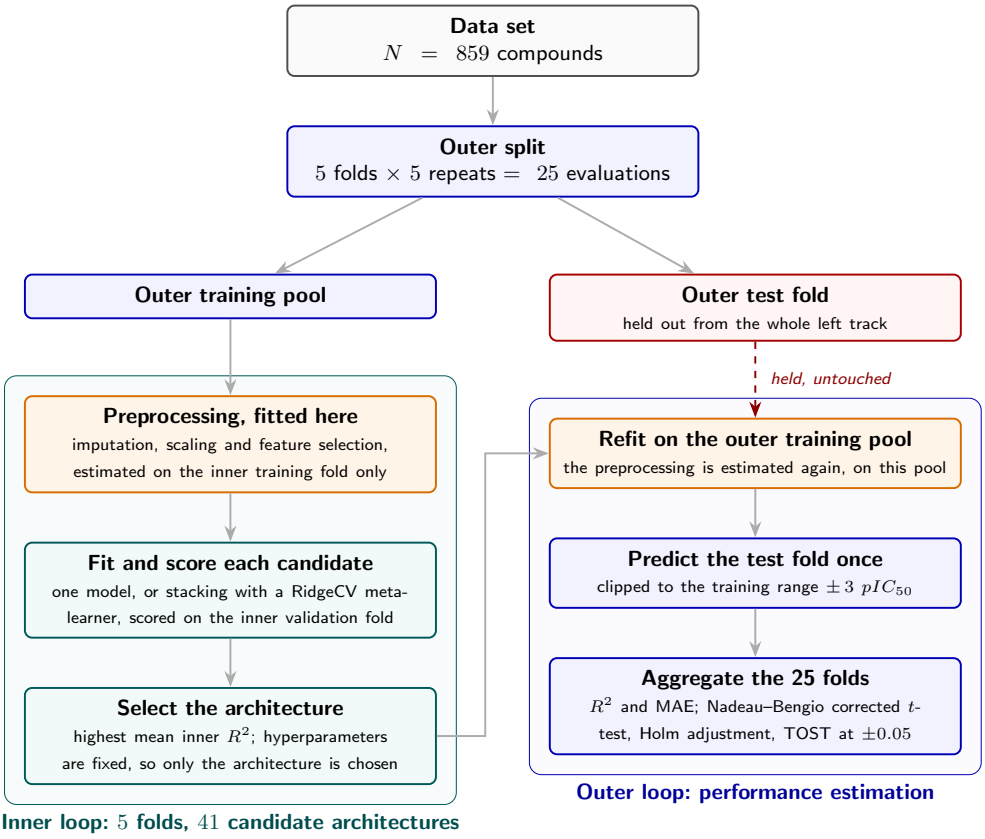


Fig. 2 One outer fold of the evaluation protocol. The split sends the training pool down the left track and the test fold down the right one. The inner loop selects an architecture among the 41 candidates without ever seeing the test fold, which is held untouched; only the chosen architecture crosses to the right track, where it is refit and scored once. The two orange boxes are the reason the estimate is leakage-free: imputation, scaling and supervised feature selection are not computed once on the whole data set but estimated again inside each partition they are used in, first on the inner training fold during selection and then on the outer training pool for the final fit, and applied to held-out data without re-estimation. Because the 25 outer partitions are shared by every representation and every model, all comparisons in this study are paired on the same molecules.

training pool it evaluated all 41 single, pairwise and three-way combinations of six regressor families (Random Forest, Extra Trees, LightGBM, XGBoost, SVR with an RBF kernel and $C = 10$, and k-nearest neighbours with $k = 5$ and cosine distance; the tree ensembles used 200 trees) and kept the combination with the highest mean inner-fold R^2. All hyperparameters were fixed in advance, so the inner loop selects among architectures only; multi-model combinations were combined by a `StackingRegressor` with a `RidgeCV` meta-estimator trained on 5-fold out-of-fold base predictions. Fitting a global transformation before partitioning would leak data-set-wide statistics into the training phase [11, 47], so imputation, standardization and supervised feature selection were implemented as steps of the model pipeline and their parameters estimated only on the training partition at hand, an inner training fold during selection and the

outer training pool for the final fit, then applied to the held-out data without re-estimation [29]. For the continuous descriptor spaces that pipeline chained median imputation, `StandardScaler` and `SelectFromModel` keeping every descriptor with a strictly positive Random Forest importance; the binary Morgan space was passed to the regressors unchanged. The selected combination was refit on the whole outer training pool and scored once on the held-out outer fold, with predictions clipped to the training range extended by 3 pIC_{50} units as a safeguard.

Repeated cross-validation violates the independence assumption of standard parametric tests, because its training sets overlap across iterations. Every comparison between models was therefore made on the shared outer folds with the Nadeau–Bengio corrected paired t-test [48], which inflates the variance of the mean difference over the $J = 25$ outer folds from $\hat{\sigma}^2/J$ to $(1/J + n_{test}/n_{train})\,\hat{\sigma}^2$, with $n_{test}/n_{train} = 1/4$ for the 5-fold outer loop; p values within each family of comparisons were then adjusted by Holm's step-down procedure [49]. Because a non-significant difference is not evidence that two models perform alike, each comparison was additionally submitted to two one-sided tests [50] against an equivalence margin of ± 0.05 in R^2 and in MAE, equivalence being declared only when the 90% confidence interval of the difference lies entirely inside that margin. The margin is our own choice, as the literature offers no standard value for it, and a wider one would make equivalence easier to declare. Confidence intervals for the hold-out ablation of Section 3.2 come instead from 2000 percentile bootstrap resamples of the held-out molecules [51]; the resampled indices depend only on the seed, so the same molecules are drawn for every model and the differences are paired.

2.5 Deep learning baseline: Directed message passing neural networks

Benchmarking classical machine learning ensembles against advanced representation learning required implementing a Directed Message Passing Neural Network (D-MPNN) via the Chemprop framework (version 2.2.4) [45, 52]. To guarantee methodological parity, the D-MPNN was evaluated on the same 25 outer partitions (5×5 repeated cross-validation, random seed 42) used for the classical baselines, with each molecule predicted by a model that never saw it. Two variants were run: a default configuration, for which no inner loop was needed, and a tuned variant with a hyperparameter search nested inside each outer fold. This alignment keeps the performance metrics directly comparable and free from single-split variance artifacts.

The architecture processes molecular SMILES strings by mapping them into graph representations, designating atoms as vertices and bonds as edges [53]. Initial feature vectors capturing atomic identity and topological connectivity were constructed using `SimpleMoleculeMolGraphFeaturizer()`. The network applied `BondMessagePassing()` (hidden size 300, depth 3, no dropout) to extract embeddings directly from directed edges rather than vertices. A `MeanAggregation()` function pooled these atomic embeddings into a unified molecular vector.

Within each training partition, the continuous pIC_{50} targets were standardized with the training mean and standard deviation, and predictions were transformed back to the original scale. The pooled vector was batch-normalized and fed into a

`RegressionFFN()` (one hidden layer of 300 units) to yield the final predictive output. The mean squared error on the standardized targets was the training loss. Model weights were optimized via PyTorch Lightning for a fixed number of 100 epochs on CPU, using Chemprop's default learning-rate schedule (2 warm-up epochs, initial rate 10^{-4}, peak 10^{-3}, final 10^{-4}) and a per-fold random seed (42 plus the fold index). No early stopping or checkpoint selection was applied, so the model after the last epoch was used and the held-out fold never influenced training.

Unlike the classical ensembles, whose architecture was selected by an inner search, the default D-MPNN used Chemprop's default hyperparameters (hidden size 300, depth 3, no dropout). A search matching the classical one would require 205 trainings per outer fold (41 architecture combinations across 5 inner splits), which at the measured cost of one training is not feasible for the full protocol (Section 3.3); the tuned variant therefore uses a reduced search.

The tuned variant searched a grid of 8 configurations (hidden size 300 or 600, message-passing depth 3 or 5, dropout 0 or 0.2; the default configuration is one of them). Inside each outer training pool, 20% of the compounds were held out at random, every configuration was trained on the remaining 80% and the one with the lowest root-mean-square error on the held-out part was retained. That configuration was then retrained on the whole outer training pool, with the same 100-epoch schedule and per-fold seed, and scored once on the outer test fold. The held-out outer fold therefore played no role in the choice. The search is deliberately smaller than the classical one (a single inner split instead of 5 folds, and 8 configurations instead of 41 combinations), so it tests whether a moderate amount of tuning helps rather than the best result that exhaustive tuning could reach. The default variant isolates the intrinsic inductive bias of the graph architecture from the confounding artifacts of small-sample tuning [2, 11], and the tuned variant checks whether that choice handicapped the network.

The default and the tuned variant both test one architecture on a modest budget. To separate what the graph architecture contributes from what the training budget and external data contribute, four further neural models were evaluated on the same 25 outer folds, each fitted on the outer training partition alone and scored exactly like the models above. The first is a feed-forward network on the same 2048-bit ECFP4 fingerprints used by the classical ensembles, which keeps their input and learns its own transformation of it. The second concatenates the vector learned by the D-MPNN with the RDKit 2D descriptors before the feed-forward head, so that the network receives the descriptor block as well. The third averages the predictions of five D-MPNN that differ only in their random seed, which buys accuracy with computation alone. The fourth initializes the message-passing weights from CheMeleon [54], a model pretrained on classical molecular descriptors computed for a large unlabeled corpus, and is the only model in this study that sees information from outside the benchmark. None of the four was tuned.

Finally, the architecture of the original study was re-evaluated without its leaks on the same 25 outer folds, in two variants: the deep ensemble on all descriptors (DNN baseline) and the complete original pipeline of importance-based selection and Gaussian augmentation (DNN full), both with the selection and the augmentation confined to each training partition and with early stopping on an internal split. This

separates what the architecture contributes from what the protocol contributes, and places the original model on the same folds as every other model in this study.

2.6 Application: Screening within the Applicability Domain

The feature-selected Morgan-fingerprint stacking ensemble of the ablation above (Random Forest, XGBoost and SVR combined by a ridge meta-learner, trained without augmentation on the 730-compound development partition of the 15% hold-out split; hold-out $R^2 = 0.748$) was applied to a library of FDA-approved compounds ($N = 3378$). Structural pre-filtering kept only compounds containing H, C, N, O, F, P, S, Cl, Br and I, a set that covers all but 0 of the training compounds, and a size ceiling (molecular weight ≤ 1000 Da; the largest training compound weighs 594 Da) removed molecules whose potency the model could not meaningfully extrapolate.

Prediction on a library of this kind is only meaningful where the model has support. Following the finding of Sheridan et al. that prediction error grows as a molecule becomes less similar to the training set [55], every FDA compound was given its mean Tanimoto similarity (2048-bit Morgan fingerprints) to its 5 most similar training compounds, and was flagged as inside the applicability domain when this value was at least 0.638, the threshold obtained for the training compounds themselves as the mean plus 0.5 standard deviations of their distances [56]. The multiplier is an arbitrary choice: the literature offers no universal rule, and a comparison of applicability-domain methods concludes that the appropriate one has to be determined for the data set at hand [44]. For that reason no compound was excluded on this flag, which is reported with every candidate.

Candidates were ranked by predicted potency alone. Four ligand efficiency metrics were computed from the prediction and reported beside it, to show how much of a predicted potency is accounted for by molecular size [57, 58]; none of them was used as a filter, and their definitions and the assumptions they carry are given in the Supplementary Material. The ranked candidates were then submitted to an orthogonal structure-based check with AutoDock Vina against the TgDHFR crystal structure 6AOG [59, 60], whose protocol, validated by re-docking the co-crystallized ligand to a heavy-atom RMSD of 1.11 Å, is described in the Supplementary Material together with the consensus thresholds.

This stage generates computational hypotheses. None of the candidates was tested experimentally, and selectivity against human DHFR, which the antifolates among them are known to inhibit, was not assessed.

2.7 Computational Environment and Reproducibility

To support algorithmic transparency and computational reproducibility [32], all data preparation, model training, and evaluation steps were programmatically implemented in Python (version 3.11.16) and chained in a single entry point that runs every step of the study in order, from the construction of the training sets to the final consistency check of the manuscript. The computational environment relied on a standardized core software stack:

- **Cheminformatics and Featurization:** RDKit (version 2025.9.3) [36] executed all molecular parsing, structural standardization, and descriptor generation (2D, 3D, and ECFP4); Open Babel (version 3.1.1) [61] converted the receptor to PDBQT.
- **Classical Machine Learning:** The nested cross-validation framework [8, 43] and baseline algorithms leveraged `scikit-learn` (version 1.6.0) [62], `xgboost` (version 3.1.2) [63], and `lightgbm` (version 4.6.0) [64]. Data manipulation, matrix operations, and statistical tests were managed via `pandas` (version 2.3.3) [65], `numpy` (version 1.26.4) [66], and `scipy` (version 1.16.3).
- **Deep Learning Infrastructure:** The D-MPNN graph models utilized `chemprop` (version 2.2.4) [52, 53] and `lightning` (version 2.6.5), with underlying tensor operations supported by PyTorch (`torch`, version 2.5.1) [67].
- **Docking:** AutoDock Vina (Python bindings, version 1.2.7) [59] performed the docking and ligands were prepared with Meeko (version 0.8.0).
- **Hardware Specifications:** All model training, architecture selection, and evaluation protocols ran on an Ubuntu 26.04 LTS workstation with two Intel(R) Xeon(R) E5-2695 v2 processors (24 physical cores, 48 threads, 2.4 GHz) and 247 GB of RAM, using CPU only; no GPU was involved.

All random number generators were seeded (seed 42). The models of the augmentation study were fitted single-threaded, because a multi-threaded random forest sums its trees in a thread-dependent order and changes the last bits of its predictions; parallelism was applied across independent tasks (features in the permutation importance, runs in the ensemble) so that the results do not depend on the number of workers. Together with the numerical tolerance of the selection rule (Section 2.1), this made the feature selection reproducible: permutation importances computed with 46 and with 8 workers were bitwise identical. The graph-model benchmark, in contrast, is not bit-reproducible: training the network on several CPU threads sums floating-point terms in a thread-dependent order, which a fixed seed does not control. Three executions of the same code on the same outer folds differed by 0.004 in mean fold R^2, well inside the confidence interval of the estimate itself ([0.678, 0.772]), so the variation affects the third decimal of the neural results and none of the comparisons reported here. Every classical result, by contrast, reproduces exactly, because the outer folds come from the fixed seed and both the preprocessing and the architecture search are confined to each training fold. The wall-clock time of every step was recorded on this machine, without normalizing by thread count, and is reported in Section 3.3.

3 Results

3.1 The Reported Gain and Where It Comes From

Run unmodified, the published code reproduced its own printed results: $R^2 = 0.858$ against the printed 0.858 for the final augmented ensemble, 0.813 against 0.816 and 0.811 against 0.815 for the two feature-selected stages; only the unselected baseline differed appreciably (0.768 against 0.746). Over 10 seeds the final model gave $R^2 = 0.853 \pm 0.007$, so the published value is not the product of a fortunate seed. What follows therefore concerns the pipeline as published.

Removing its leaks one at a time accounts for most of that value (Table 1). The largest single contribution comes from the step that is hardest to see in the code: the permutation importance that selects the descriptors is computed on the test molecules, which is worth -0.048 in R^2 (95% CI [-0.073, -0.013]). Guiding early stopping with the test set adds -0.026 ([-0.042, -0.011]), and re-partitioning the data after augmentation, which lets a noisy copy of a test molecule train the model, adds -0.028 ± 0.007. This last figure carries no interval because the published protocol scores itself on the partition its own re-splitting produces, so it shares no test molecules with the rungs below it and the difference cannot be paired; it is reported over seeds instead. Each of these three leaks lowers the score in every one of the 10 seeds (Table S1 of the Supplementary Material). Removing duplicate structures from both sides of the split costs a further -0.011, the only step whose interval contains zero ([-0.026, $+0.005$]).

Table 1 Performance of the original pipeline as each information leak is removed, over 10 seeds. ΔR^2 is the change with respect to the rung above, with its 95% bootstrap confidence interval over the 130 molecules common to rungs R1–R4. R0 builds its own evaluation partition by re-splitting the augmented data, so it shares no test molecules with the rungs below and the difference of R1, which is measured against it, cannot be paired: [a] that one entry is given as the mean over seeds with its standard deviation instead of a bootstrap interval. The last row is a model with no feature selection, fitted on the same clean split; [b] its entry is the difference of R4 with respect to it, not a step of the ladder.

Rung	**Leak removed**	**Features**	R^2	ΔR^2
R0	None: the protocol as published	941	0.853 ± 0.007	
R1	Synthetic copies of test molecules reaching the training set	941	0.825 ± 0.005	-0.028 ± 0.007[a]
R2	Early stopping and model choice guided by the test set	945	0.800 ± 0.013	-0.026 [-0.042, -0.011]
R3	Feature importance computed on the test set	385	0.752 ± 0.019	-0.048 [-0.073, -0.013]
R4	The same structure on both sides of the split	330	0.740 ± 0.024	-0.011 [-0.026, $+0.005$]
Clean	Reference: no feature selection at all	1858	0.752 ± 0.008	-0.012 [-0.025, $+0.030$][b]

Together the four corrections take the model from $R^2 = 0.853$ to 0.740, a drop of -0.113 ± 0.023. The end point cannot be distinguished from a model with no feature selection at all (0.752 ± 0.008; difference -0.012, 95% CI [-0.025, $+0.030$]): once the importance is computed honestly, the selection buys nothing. That level is also the one reported by the original study for its own unselected baseline (0.75) and the one the classical ensembles reach in Section 3.2.

The evaluation partition itself shows the mechanism most directly. Because the published protocol augments the data and then splits it again, the partition it scores as a test set holds 131 molecules, of which 111 come from the original training set and only 20 are molecules the model had not already seen.

The same mechanism is visible in our own classical replication, with no neural network involved. Computing the permutation importance of variant B on the complete data set, as the original study does, instead of on the development partition alone, raises the hold-out R^2 from 0.748 to 0.836 ($\Delta = +0.088$, 95% CI [$+0.047$, $+0.139$]), and that of variant C from 0.732 to 0.826 ($\Delta = +0.095$, [$+0.046$, $+0.153$]). The two procedures do not even select the same features: of the bits retained by either of them, only 32.0% are retained by both (Jaccard index 0.3197), and the leaky selection keeps 446 bits against the 747 of the clean one. Seeing the test molecules does

not merely sharpen the choice; it produces a different model. Table S2 of the Supplementary Material gives the five variants of this replication side by side. A leaky selection on a classical learner thus reaches 0.836, close to the 0.82 reported for the feature-selected model of the original study, while the same selection performed correctly reaches 0.748. The gain credited to feature selection is reproducible only when the selection is allowed to see the test data.

3.2 Comparative Performance of Classical ML Across Feature Spaces

We established a baseline for TgDHFR inhibitor prediction by evaluating classical machine learning ensembles across four molecular representations with a 25-fold nested cross-validation framework. Figure 3 shows the distribution of the 25 outer-fold R^2 values of each, in the order used below. The purely structural Morgan fingerprints (ECFP4, 2048 bits) reached $R^2 = 0.769 \pm 0.041$ (MAE $= 0.434$), the 3D hybrid space $R^2 = 0.774 \pm 0.041$ (MAE $= 0.428$), and RDKit 2D descriptors combined with fingerprints $R^2 = 0.774 \pm 0.043$ (MAE $= 0.428$). RDKit 2D descriptors alone were clearly inferior ($R^2 = 0.728 \pm 0.048$, MAE $= 0.468$). The box plot prints each mean and standard deviation under its label, rounded to three decimals, and shows R^2 only; Figure S1 of the Supplementary Material gives the per-fold values of both metrics with their confidence intervals.

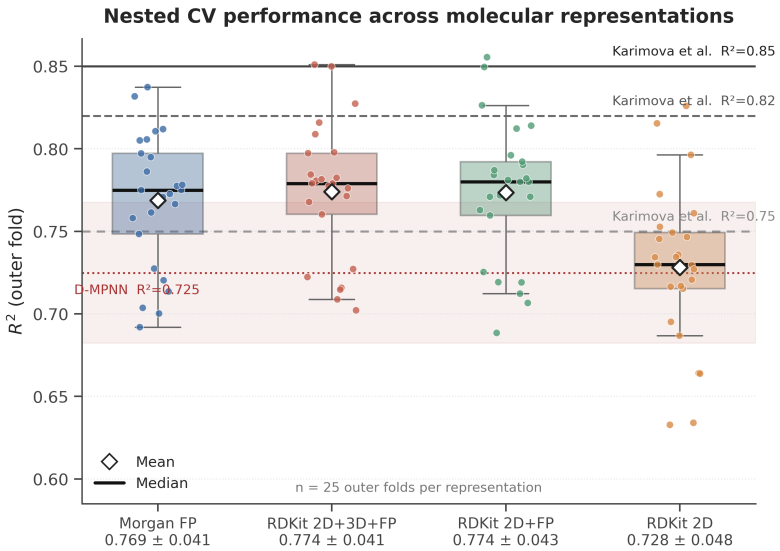


Fig. 3 Nested cross-validation R^2 across four molecular representations (each point is one of the 25 outer folds). Horizontal lines mark the three values reported in the original study [25]; the dotted line and band show the mean ± SD of the D-MPNN benchmarked in Section 3.3. The three representations that contain fingerprints are statistically indistinguishable in R^2.

Differences between representations were assessed with Nadeau-Bengio corrected paired t-tests on the shared outer folds. The three fingerprint-containing spaces could not be separated in R^2 (Morgan versus 2D descriptors with fingerprints, $p = 0.5061$; Morgan versus the 3D hybrid, $p = 0.4006$; 2D descriptors with fingerprints versus the 3D hybrid, $p = 0.8243$). In MAE, Morgan trailed the two hybrid spaces by 0.006 and 0.006 pIC_{50} units ($p = 0.2739$ and $p = 0.2563$), a small gap that is no longer significant after Holm adjustment for the six comparisons ($p_{\mathrm{Holm}} = 0.7689$ for both). Table S3 of the Supplementary Material gives the six comparisons in both metrics. In contrast, 2D descriptors alone were significantly worse than every space that includes fingerprints (versus 2D descriptors with fingerprints, $p = 0.0002$; versus the 3D hybrid, $p = 0.0001$; versus Morgan, $p = 0.0020$). Topological fingerprints are therefore the primary driver of predictive capacity, and the ten 3D shape descriptors derived from ETKDGv3 embedding and MMFF94 minimization added no measurable advantage for this benchmark.

To rule out chance correlation, the y-randomization tests prescribed by Rücker et al. [16] were run in four forms, each with 100 randomizations scored by 5-fold cross-validation (Table S4). Applied to the stacking architecture alone (RF+XGB+SVM: random forest, XGBoost and SVR with a ridge meta-learner, on all 2048 bits), the real model reached $R^2 = 0.779$ (a single 5-fold partition, which explains the small difference from the nested estimate) while the permuted models collapsed to -0.014 ± 0.008, a separation of $Z = 93.9$ standard deviations. The demanding form of the test repeats the whole procedure that produces the deployed screening model, bit selection included, inside every randomization: the separation then falls to $Z = 47.8$ when the response is permuted and to $Z = 27.0$ when the descriptors themselves are replaced by random ones, the smallest of the four and still far outside the randomized distribution. All four reach the smallest p attainable with 100 permutations ($p = 0.0099$ against $\alpha = 0.01$). Neither the architecture nor the selection procedure therefore owes its accuracy to chance correlation.

Architecture selection was only partly stable, and markedly less so outside the Morgan space (Figure 4; Figure S2 of the Supplementary Material gives the full grid of combinations chosen in every fold and representation). Within the Morgan space, Support Vector Regression entered the selected ensemble in 100% of the outer folds, XGBoost in 92% and Random Forest in 84%, followed at a distance by k-nearest neighbors (12%), LightGBM (4%) and Extra Trees (4%); the selected ensembles contained one, two, or three models in 0%, 4%, and 96% of the folds. One combination dominated without being universal: RF+XGB+SVM won 76.0% of the folds (19/25), followed by XGB+SVM+kNN (3/25) and 3 further architectures, none of them chosen in more than 1 of the 25 folds. The descriptor-based spaces were less settled. Their base learners are consistent (Extra Trees entered in 100%, 100%, and 100% of the folds of the 2D, 2D+FP, and 2D+3D+FP spaces, and SVR in 92%, 96%, and 100%), but the exact three-model combination varied much more: the most frequent one was chosen in 52.0%, 32.0%, and 60.0% of the folds, against 76.0% for Morgan. Table S5 reports the composition of the selected ensembles for every representation. Aggregated performance is stable across outer folds, but outside the Morgan space the 859-compound benchmark does not let the inner loop settle on one ensemble, so

the reported performance is that of the selection procedure rather than that of a fixed architecture.

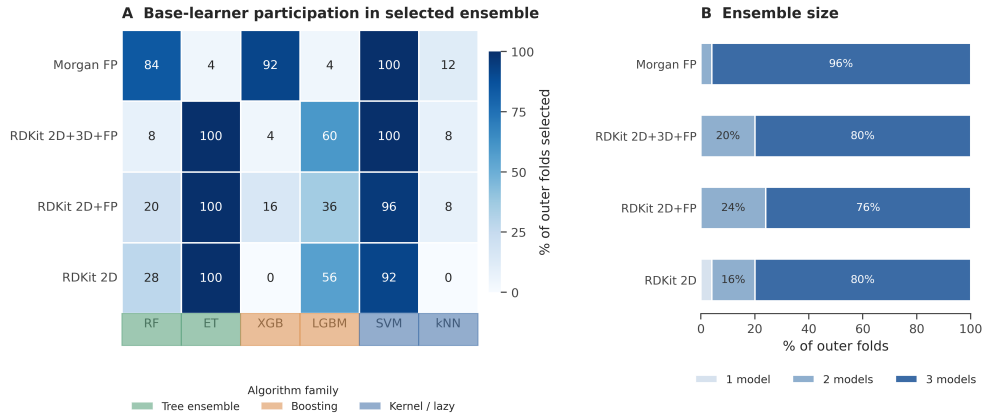


Fig. 4 Composition of the stacking ensembles selected by the inner loop in the 25 outer folds. (A) Percentage of folds in which each base learner entered the selected ensemble, per representation. (B) Distribution of ensemble size (one, two, or three base models). No single architecture dominates in the Morgan space.

We then placed these results against the incremental progression reported in the original study (Table 2). Our classical stacking ensemble trained solely on Morgan features ($R^2 = 0.769$) performed comparably to their unselected 2D/3D/FP baseline ($R^2 = 0.75$); under a Nadeau-Bengio corrected one-sample t-test the difference was not significant ($t = 0.862$, $p = 0.3971$). In contrast, the literature value for the model with permutation-importance feature selection ($R^2 = 0.82$) lies significantly above our nested cross-validated Morgan estimate ($t = -2.339$, $p = 0.0280$), and so does the final data-augmented deep neural network ensemble ($R^2 = 0.85$; $t = -3.712$, $p = 0.0011$). This comparison has a limitation: the original values come from a single hold-out split without a reported dispersion and are treated here as fixed constants, so the test does not capture the sampling variability of that split. Achieving a competitive baseline ($R^2 \approx 0.75$) does not require computationally expensive 3D descriptors or complex neural network architectures; simple 2D fingerprints paired with classical algorithms suffice.

To test whether the interventions credited with the gains of the original study transfer to classical learners, we replicated its protocol (15% hold-out, seed 42; 129 molecules held out before any selection or augmentation, 730 for development) on a stacking ensemble (Figure 5). The unselected baseline reached $R^2 = 0.733$ (MAE $= 0.441$). Permutation-importance selection (747 of 2048 bits retained) yielded $R^2 = 0.748$ (MAE $= 0.433$), and Gaussian-noise augmentation yielded $R^2 = 0.732$ (MAE $= 0.439$). Paired bootstrap differences showed no significant effect of either intervention: selection changed R^2 by $+0.015$ (95% CI $[-0.013, +0.043]$) and augmentation by -0.001 (95% CI $[-0.027, +0.020]$) relative to the baseline, and both intervals include zero.

Table 2 Methodological comparison isolating the performance impact of architectural complexity, descriptor dimensionality, and data augmentation in TgDHFR bioactivity prediction.

Model Framework	Descriptors	Data Augmentation	Test R^2 (mean ± std)
Current Study (Classical Ensemble)	ECFP4 (2D)	None	0.769± 0.041
Baseline [25]	2D/3D/FP	None	0.75[a]
Feature-selected model [25]	2D/3D/FP (Selected)	None	0.82[a]
Augmented Final Model [25]	2D/3D/FP (Selected)	Gaussian Noise	0.85[a]

[a] Standard deviation not reported (NR) in the original study.

The hold-out intervals are wide, spanning roughly 0.2 R^2 units, and contain the original baseline of 0.75; the original values of 0.82 and 0.85 lie above the upper limits of the intervals for selection ($p_{\text{boot}} = 0.0850$) and augmentation ($p_{\text{boot}} = 0.0050$), respectively. The gains reported for the original deep ensemble were therefore not reproduced by feature selection or augmentation applied to a classical learner; they may depend on the neural architecture, or reflect the sampling variability of a single hold-out split. Two properties of the replicated interventions limit how strong this negative result is. First, the permutation importance was computed on the data the model was fitted on, so the selection removes essentially the bits the model never uses and little else, and no change is expected for tree-based learners. Second, Gaussian noise of $\sigma \leq 0.01$ added to binary fingerprint bits barely moves the split thresholds of tree models, so the augmented copies are almost identical to the originals. Neither intervention was therefore a demanding test of its idea.

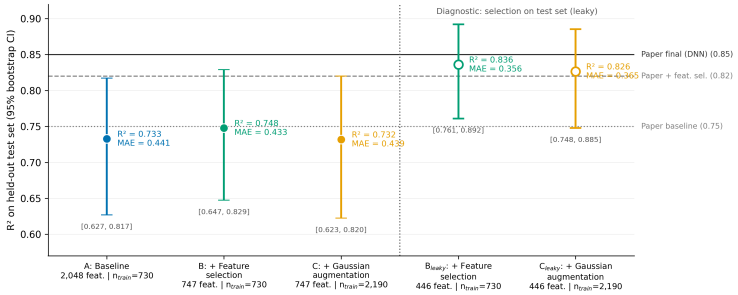


Fig. 5 Replication of the original hold-out protocol on a classical stacking ensemble (129 held-out molecules). Points are R^2 on the test set with 95% bootstrap confidence intervals for the baseline (A), permutation-importance feature selection (B), and feature selection plus Gaussian-noise augmentation (C); horizontal lines mark the values reported in the original study [25].

3.3 Benchmarking Tree-Based Ensembles Against Graph Neural Networks

Graph neural networks learn molecular representations directly from the atom–bond graph, but their advantage is expected to shrink when labeled data are scarce. To test whether representation learning helps on the TgDHFR dataset (859 compounds), a Directed Message Passing Neural Network (D-MPNN, Chemprop) was evaluated on exactly the same 25 outer folds as the classical ensembles, in two variants described in the Methods: default hyperparameters, and a hyperparameter search nested inside each outer fold.

The default D-MPNN reached a mean fold R^2 of 0.725 ± 0.043 and a mean absolute error of 0.474, compared with $R^2 = 0.769 \pm 0.041$ for the Morgan-fingerprint stacking ensemble (Figure 6). Because the folds are shared, the models can be compared with the same Nadeau–Bengio corrected paired t-test used for the classical representations. The network was numerically below the Morgan ensemble ($\Delta R^2 = -0.044$) and the fingerprint-based hybrid ensembles ($\Delta R^2 = -0.049$ and -0.049 against 2D+FP and 2D+3D+FP), and on par with the 2D descriptors alone ($\Delta R^2 = -0.003$, Holm-adjusted $p = 0.866$). After Holm adjustment the three deficits against fingerprint-based ensembles were significant, but only just: the smallest adjusted p was 0.048 for R^2 and 0.019 for MAE, so the evidence sits immediately below the conventional threshold and should be read as a small, consistent shortfall rather than a decisive one.

Tuning did not change this picture. In the nested search, the selected configuration varied across folds (7 of the 8 configurations were chosen at least once, and the default one in only 4% of the folds; Table S6 of the Supplementary Material gives the configuration selected in each fold with its inner error), but the tuned network reached $R^2 = 0.726 \pm 0.034$ and MAE = 0.470, indistinguishable from the default one ($\Delta R^2 = +0.001$, $p = 0.930$; ΔMAE $= -0.004$, $p = 0.713$). Against the classical ensembles the tuned network was again numerically lower than Morgan ($\Delta R^2 = -0.043$, Holm-adjusted $p = 0.044$) and the two fingerprint-based hybrids ($\Delta R^2 = -0.047$ and -0.048), and on par with the 2D descriptors ($\Delta R^2 = -0.002$, $p = 0.925$). As for the default network, the deficits against the fingerprint-based ensembles were significant by a narrow margin (smallest adjusted p: 0.041 for R^2 and 0.023 for MAE), and the comparison with the 2D descriptors alone was not.

Equivalence is a separate question; Table S7 of the Supplementary Material gives every comparison of the two network variants against the four representations, in both metrics, from which the figures below are taken. With the ± 0.05 margin, the tuned network was equivalent to the 2D descriptors in R^2 (90% CI $[-0.037, +0.033]$, $p = 0.013$), but equivalence to the Morgan ensemble could not be shown for R^2 (90% CI $[-0.071, -0.015]$, $p = 0.330$). Equivalence in MAE could not be shown either ($\Delta = +0.036$, 90% CI $[+0.011, +0.062]$, $p = 0.182$), because the upper limit of the interval exceeds the margin. The two tests therefore agree: against the Morgan ensemble the network is neither shown to be equivalent nor comfortably shown to be worse, and the data leave open a deficit in R^2 as large as $\Delta R^2 = -0.071$. Against the 2D descriptors, by contrast, equivalence holds in both metrics ($p = 0.013$ for R^2 and 0.002 for MAE). The tuned search was deliberately modest (Methods), so a larger search or a different architecture could change this.

Both graph-network variants and the classical ensembles fall in the same range as the original study's unselected baseline ($R^2 = 0.75$). The final augmented ensemble of the original study ($R^2 = 0.85$) lies above this range, but, as discussed in Section 3.2, that value comes from a single hold-out split and was not reproduced by selection or augmentation on a classical learner. Under the data volume of this benchmark, the graph architecture therefore did not provide a gain over fixed fingerprints; it trailed the fingerprint-based ensembles by about -0.043 in R^2, a difference whose adjusted p values fall just below 0.05 and which leaves the network on a par with 2D descriptors alone. The defensible reading is that the graph representation, trained this way, buys nothing here, not that it is inferior in principle.

Computational cost does not favor the classical approach in this protocol, but the comparison needs care (Table 4). One default D-MPNN training took about 179 s on the workstation CPU (about 75 minutes for the 25 folds), whereas a classical outer fold with its complete inner search took between 1182 and 2586 s depending on the representation (1987 s, about 13.8 h in total, for Morgan). The default network received no model selection. When it did, in the tuned variant, one outer fold took 2192 s, about 15.2 h of worker time for the whole benchmark. That is more than the 13.8 h of the Morgan protocol, although not more than the most expensive classical representation (2586 s per fold), and the network's search was the smaller of the two. The case for the classical ensembles therefore rests on accuracy that is not lower, on simpler inputs and models that run on a CPU, and on a deployed model that uses only a subset of the fingerprint bits, and not on a shorter benchmark. For the study as a whole, a complete execution from scratch takes on the order of eight days on this workstation, dominated by the three nested cross-validation runs (main, sensitivity and scaffold-grouped, between 46 and 57 hours each) and the y-randomization test (about 19 hours); the repository documents the measured cost of every step.

The gap to the fingerprint ensembles is a property of this training budget rather than of the graph architecture (Table 3). Averaging five D-MPNN over their seeds recovers part of it, to $R^2 = 0.741 \pm 0.040$, and leaves a difference from the Morgan ensemble that no longer reaches significance ($\Delta R^2 = -0.028$, Holm-adjusted $p = 0.162$) although equivalence is still not established in R^2 ($p = 0.058$). Initializing the same network from the CheMeleon pretrained model closes the gap altogether: $R^2 = 0.761 \pm 0.041$ and MAE 0.442, within -0.007 of the Morgan ensemble, and the only model of this study that is statistically equivalent to it in both metrics ($p = 0.004$ for R^2 and 0.007 for MAE). Table S8 gives each of these models against each representation in both metrics. The two models that add information of a different kind did not help: giving the D-MPNN the RDKit descriptors lowered it to 0.701, and a feed-forward network on the same fingerprints the classical ensembles use reached 0.709.

Parity therefore costs something. The pretrained network needed 270.0 minutes per outer fold against the 1987 s of a complete classical fold with its architecture search, roughly eight times as much computation, and it reaches the accuracy of the fingerprint ensemble without exceeding it. The seed ensemble is the more economical of the two routes, at 34.6 minutes per fold. Read together with the default and tuned

Table 3 Neural models evaluated on the 25 shared outer folds, ordered by R^2, with the Morgan ensemble and the default D-MPNN as references. ΔR^2 is the difference from the Morgan ensemble, p its Holm-adjusted value within the four classical representations, and p_{equiv} the two one-sided tests against the ± 0.05 margin. Times are the mean wall-clock time of one outer fold and are not normalized by thread count.

Model	R^2	MAE	**min/fold**	ΔR^2	p	p_{equiv}
Morgan ensemble (reference)	0.769 ± 0.041	0.434	33.1	—	—	—
D-MPNN, default	0.725 ± 0.043	0.474	3.0	—	—	—
D-MPNN initialized from CheMeleon	0.761 ± 0.041	0.442	270.0	-0.007	1.000	0.004
Ensemble of 5 D-MPNN	0.741 ± 0.040	0.460	34.6	-0.028	0.162	0.058
MLP on ECFP4	0.709 ± 0.054	0.499	0.5	-0.060	0.097	0.646
D-MPNN with RDKit 2D descriptors	0.701 ± 0.050	0.495	8.1	-0.068	0.012	0.805

variants, these results place the ceiling of representation learning on this data set at the level of a fixed fingerprint, reached only with pretraining on an external corpus.

The architecture of the original study behaves the same way once its protocol is repaired. Evaluated on these folds with the selection and the augmentation confined to each training partition (Section 2.2), the deep ensemble reached $R^2 = 0.673 \pm 0.086$ on all descriptors and 0.715 ± 0.077 with the selection and the augmentation of the original pipeline, a difference of $+0.042$ that the data cannot resolve ($p = 0.414$). Both sit below the Morgan ensemble, by -0.054 for the better of the two, with a confidence interval ($[-0.131, +0.023]$) too wide to settle the comparison. The interventions credited with the reported gain therefore do not reproduce on the architecture they were proposed for either, once the test molecules are kept out of them.

Table 4 Run time and accuracy of every model. Time is the mean wall-clock time of one outer fold on the workstation, including the inner search where one exists, and the 25-fold total is that time multiplied by 25. Times are not normalized by thread count (the classical models used all threads on one fold at a time; the tuned D-MPNN ran two training processes in parallel).

Model	**Time per outer fold (s)**	**25 folds (h)**	**Model selection**	R^2
Morgan FP	1987	13.8	Yes (41 combinations)	0.769 ± 0.041
RDKit 2D+FP	2504	17.4	Yes (41 combinations)	0.774 ± 0.043
RDKit 2D+3D+FP	2586	18.0	Yes (41 combinations)	0.774 ± 0.041
RDKit 2D	1182	8.2	Yes (41 combinations)	0.728 ± 0.048
D-MPNN, default	179	1.2	No	0.725 ± 0.043
D-MPNN, tuned	2192	15.2	Yes (8 configurations)	0.726 ± 0.034

3.4 Robustness to Scaffold-Grouped Splits

Random folds place close analogs of a test compound in the training set, which favors models that memorize local structure. To measure how much of the reported accuracy depends on this, every model of this study was re-evaluated with scaffold-grouped

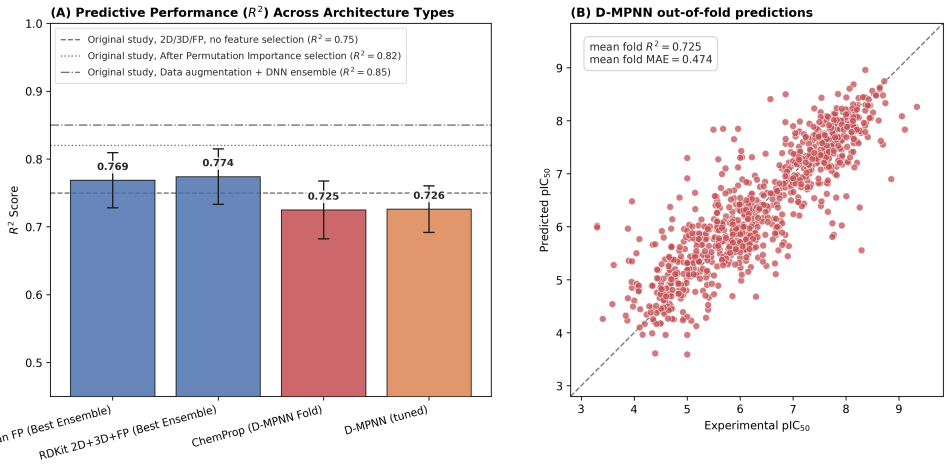


Fig. 6 (A) Mean R^2 over the 25 shared outer folds for the best classical ensembles and the D-MPNN with default and tuned hyperparameters; error bars are the standard deviation across folds, and horizontal lines mark the values reported in the original study. (B) Default D-MPNN predicted versus experimental pIC$_{50}$ for every compound, averaged over the five repeats of the cross-validation (each prediction comes from a model that did not see the compound).

outer folds (Section 2.4), keeping the 25 folds, the hyperparameters and the shared partitions. Table 5 summarizes the outcome.

Table 5 Cross-validated performance (mean ± SD over 25 outer folds) under random and scaffold-grouped splits, ordered by the size of the drop in mean R^2. Both protocols share their outer partitions across models, so every column is a paired comparison.

		R^2			
Model	**Family**	Random	Scaffold	**Drop**	**MAE (scaffold)**
RDKit 2D ensemble	classical	0.728 ± 0.048	0.492 ± 0.116	0.236	0.673
DNN with selection and augmentation	neural	0.715 ± 0.077	0.510 ± 0.127	0.205	0.611
D-MPNN, default	neural	0.725 ± 0.043	0.521 ± 0.127	0.204	0.647
RDKit 2D+FP ensemble	classical	0.774 ± 0.043	0.571 ± 0.104	0.203	0.613
RDKit 2D+3D+FP ensemble	classical	0.774 ± 0.041	0.574 ± 0.106	0.200	0.612
Ensemble of D-MPNN	neural	0.741 ± 0.040	0.555 ± 0.113	0.186	0.623
DNN baseline	neural	0.673 ± 0.086	0.489 ± 0.169	0.185	0.614
Morgan ensemble	classical	0.769 ± 0.041	0.594 ± 0.097	0.175	0.598
D-MPNN initialized from CheMeleon	neural	0.761 ± 0.041	0.601 ± 0.087	0.161	0.599

Holding scaffolds out costs every model between 0.161 and 0.236 units of R^2, with the Morgan ensemble falling from 0.769 to 0.594 (MAE from 0.434 to 0.598) and the default D-MPNN from 0.725 to 0.521 (MAE from 0.474 to 0.647). The fold-to-fold spread also more than doubled (SD of R^2 from 0.041 to 0.097 and from 0.043 to 0.127, respectively), because the difficulty of a fold now depends on which scaffolds it contains. The random-split estimates of the main benchmark therefore overstate the performance expected on new chemotypes.

The size of the drop is not random across models. The two representations that carry the least information beyond the training set lose the most (0.236 for the 2D descriptors), and the two that carry the most lose the least: the Morgan ensemble (0.175) and the pretrained network (0.161), which ends as the highest-scoring model of the scaffold protocol (0.601 ± 0.087) although it cannot be separated from the Morgan ensemble there either ($\Delta R^2 = +0.007$, Holm-adjusted $p = 0.925$; equivalent in MAE, $p = 0.006$). Pretraining on an external corpus thus buys robustness to a change of chemotype rather than accuracy within one, which is the one advantage of representation learning that this benchmark does show.

The comparison between families kept its direction but not its resolution. Under scaffold splits the D-MPNN was lower than the Morgan ensemble by $\Delta R^2 = -0.073$ (corrected $p = 0.368$, Holm-adjusted; $\Delta\text{MAE} = +0.049$, $p = 0.464$). The 90% confidence interval of the R^2 difference, $[-0.144, -0.002]$, excludes zero only barely, and its lower limit extends far beyond the ± 0.05 equivalence margin; the equivalence test was accordingly not significant ($p = 0.707$). The interval carries no multiplicity adjustment while the p values do, which is why the two do not point the same way; Table S9 gives both the unadjusted and the adjusted values for every comparison under this protocol. With the present data we can thus neither claim a difference nor equivalence between the two families on new scaffolds; the sample (859 compounds) is too small to separate them given the larger variance of this protocol.

3.5 Sensitivity to Extreme Potency Values

The two records excluded by the Tukey rule (Section 2.1) were retained in a second, complete execution of the nested cross-validation benchmark ($n = 861$ versus $n = 859$ in the primary analysis; identical algorithms, search space and 5×5 protocol). If these records were informative members of the chemical series, retaining them should not degrade generalization; if they are outliers, retaining them should lower accuracy. Table 6 reports the outcome.

Table 6 Nested cross-validation performance without (primary analysis, $n = 859$) and with ($n = 861$) the two extreme-potency records. p-values compare the two runs with a Welch test using the Nadeau-Bengio variance correction [48]; the runs use different fold partitions, so folds are not paired.

	R^2				MAE			
Representation	Without	With	Δ	p	Without	With	Δ	p
RDKit 2D+3D FP	0.774	0.754	-0.020	0.52	0.428	0.440	+0.012	0.62
RDKit 2D FP	0.774	0.753	-0.020	0.53	0.428	0.439	+0.011	0.64
Morgan FP	0.769	0.750	-0.019	0.56	0.434	0.449	+0.015	0.53
RDKit 2D	0.728	0.709	-0.019	0.63	0.468	0.482	+0.014	0.61

Retaining the two records lowered R^2 in all four representations (ΔR^2 between -0.020 and -0.019) and raised the MAE (ΔMAE between $+0.011$ and $+0.015$ pIC_{50} units), a consistent direction that is nevertheless indistinguishable from fold-to-fold variability (p between 0.52 and 0.64). For the Morgan representation, R^2 moved from

0.769 to 0.750 ($p = 0.56$). The ordering of the three best representations changed between runs, but their differences were already within noise in the primary analysis, so the central conclusion (no representation is reliably superior) holds in both.

To test whether the records are predictable at all, a Morgan fingerprint Random Forest trained on the primary set was asked to predict them. The record at $pIC_{50} = 11.68$ was predicted as 6.59 (absolute error 5.09) and the record at 1.56 as 4.98 (error 3.42), that is, 8.2 and 5.5 times the out-of-fold RMSE of that model (0.62), and larger than every out-of-fold error among the 859 retained compounds.

Taken together, the two records behave as outliers with respect to the model: they cannot be anticipated from the structure–activity relationship learned from the rest of the series, and keeping them modestly worsens, without significantly altering, the reported metrics. Their exclusion is therefore justified as a pre-specified applicability-domain decision rather than as a way of improving performance, and it does not affect any comparison reported in this study. We note that the high-potency record may be a genuine measurement of a very potent antifolate rather than an error; in either case it lies outside the domain that the present data can support.

3.6 Hypothesis Generation: Screening and Docking

The deployed model, the feature-selected Morgan-fingerprint stacking ensemble of the augmentation experiment (Section 3.2), was applied to the FDA-approved library after the funnel described in the Methods: 3378 compounds were reduced to 3056 by the size and elemental filters, and all of these were scored. Their mean Tanimoto similarity to the five most similar training compounds had a median of 0.17 (5th–95th percentile 0.11–0.26), against 0.70 for the training compounds measured against each other, and only 14 of them reached the applicability-domain threshold of 0.638. The predicted potencies of this library are therefore mostly extrapolations, and each candidate's similarity to the training set is reported with its prediction; of the 17 candidates advanced to docking, 6 were inside the domain. As a consistency check, pyrimethamine was predicted at pIC_{50} 6.57 (experimental 6.56) and trimethoprim at 5.67 (experimental 5.57); both belong to the training partition, so this verifies that the final model fits its training data and says nothing about generalization.

Table S10 and Figures S3–S5 give the ranking with its efficiency metrics. The antifolates methotrexate, trimetrexate and pralatrexate lead by absolute potency, but potency scales with molecular size and their efficiency is modest (ligand efficiency between 0.27 and 0.40, against the value of 0.30 commonly taken as the lower bound of a good lead [57, 58]); methotrexate is also in the training set, so its rank partly reflects memorization. Size-corrected metrics reorder the list: triamterene, which is absent from the training data, combines a predicted pIC_{50} of 7.01 with the highest ligand efficiency (0.51) and binding efficiency index (27.7) of the table, and is therefore the clearest repurposing hypothesis of the ML stage.

Because neither ligand-based prediction nor docking is reliable alone, the 17 candidates were scored by both and classified by agreement (Figure 7), a compound being a consensus hit when the model predicts it active ($pIC_{50} > 6.5$) and it docks favorably ($\Delta G < -8.0$ kcal/mol). This yielded 4 entries (4 unique compounds): pralatrexate ($pIC_{50} = 7.0034$, $\Delta G = -8.68$ kcal/mol, the best docking score of the set),

methotrexate (7.4448, -8.66), trimetrexate glucuronate (7.9272, -8.76) and triamterene (7.0081, -8.07). Agreement between the two methods is weak: Spearman's rank correlation between predicted pIC_{50} and Vina score was $\rho = -0.313$ ($p = 0.2218$), with the expected sign but no statistical significance, and the estimate is less informative than its $N = 17$ suggests, because the four trimethoprim entries share one structure and one docking score. We therefore read the correlation only as evidence that docking does not reproduce the ML ranking.

The discordant cases bound both methods. Pyrimethamine, an established TgDHFR inhibitor, docked at only $\Delta G = -7.62$ kcal/mol and trimethoprim similarly ($\Delta G = -7.56$ kcal/mol); small ligands obtain modest Vina scores because the score grows with the number of contacts, which points to a limited sensitivity of the scoring function for small inhibitors rather than to an ML artifact. Bisacodyl is the opposite case, favored by docking ($\Delta G = -8.82$ kcal/mol) while the model predicted $pIC_{50} = 4.9528$; it carries none of the diaminopyrimidine-like features of the antifolate ligands in the training set, so a docking artifact of a flexible, hydrophobic ligand is the more plausible explanation. Without experimental data we cannot arbitrate between the two methods.

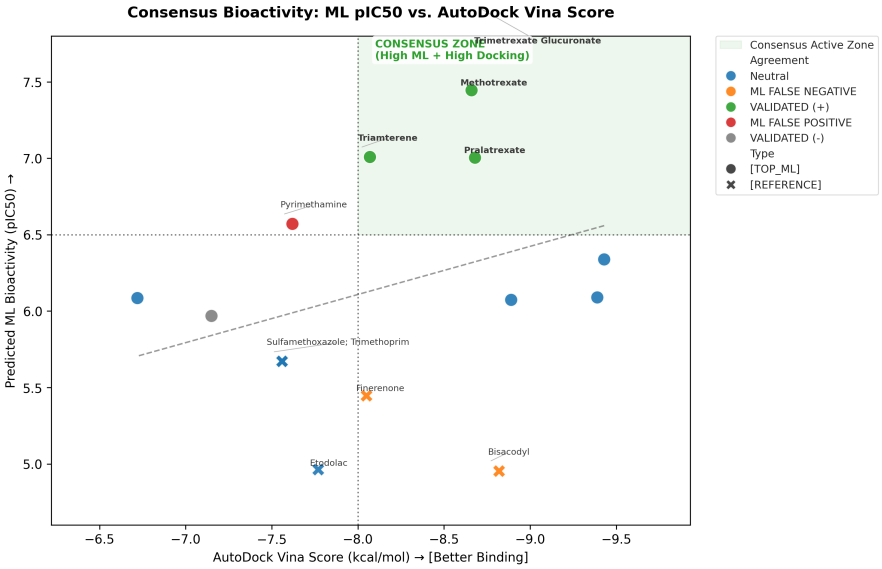


Fig. 7 Predicted pIC_{50} versus AutoDock Vina score for the 17 docked candidates, coloured by agreement class. The shaded region is the consensus zone ($pIC_{50} > 6.5$ and $\Delta G < -8.0$ kcal/mol).

This whole stage is a computational prioritization. The receptor is rigid, no water molecules or protonation states are modelled, and Vina scores are empirical estimates. Most of the library lies outside the applicability domain of the model, and selectivity against human DHFR was not assessed. The consensus hits are therefore hypotheses for experimental testing and not confirmed inhibitors.

4 Discussion

4.1 The Role of Explicit Inductive Bias in Small-Scale Molecular Datasets

Graph neural networks can learn molecular features directly from graph topology, which removes the need for hand-designed descriptors, but this flexibility is usually paid for with training data. In our benchmark, with only 859 compounds, the D-MPNN reached $R^2 = 0.725$ with default hyperparameters and 0.726 after a nested hyperparameter search, numerically below the Morgan-fingerprint ensemble ($\Delta R^2 = -0.044$ and -0.043, respectively) by margins that cleared multiple-comparison correction only narrowly, and statistically indistinguishable from the 2D descriptors alone (Section 3.3). Tuning itself changed nothing detectable ($\Delta R^2 = +0.001$), and the configuration chosen by the search varied from fold to fold. In this dataset, therefore, learning the representation brought no detectable gain over using a fixed one.

A plausible reading is that explicit chemical priors matter when labels are scarce. A circular fingerprint hands the learner a fixed, chemically motivated description of substructures, whereas a message-passing network must learn an equivalent description from a few hundred molecules and a noisy target. We stress that this is an interpretation, not a result of this study: no experiment isolates the inductive bias. The network search was modest (8 configurations, one inner split), but the gap it leaves is not an artifact of that budget. Averaging five networks over their seeds removes most of it, and initializing the message passing from a model pretrained on an external corpus removes it entirely, to the point of statistical equivalence with the Morgan ensemble in both metrics (Section 3.3). What the benchmark measures is therefore not a ceiling of the graph architecture but the price of reaching the same accuracy: parity required external data and about eight times the computation of the classical protocol, and no model we ran exceeded the fingerprint ensemble.

The augmentation experiments argue against one common remedy. The original study attributed part of its gains to Gaussian-noise augmentation and permutation-importance feature selection applied to a deep ensemble. Applied to a classical stacking ensemble under the same hold-out protocol, neither intervention improved on the unaugmented baseline (Figure 5): selection changed R^2 by $+0.015$ and augmentation by -0.001, with confidence intervals that include zero (Section 3.2). This shows that the gains of the original study did not transfer to a classical learner under these implementations, which are weak tests for tree-based models (the selection only removes unused bits and the noise is negligible for binary inputs; Section 3.2). The audit of Section 3.1 explains where those gains come from: removing the leaks of the published code one at a time lowers its R^2 to the level of a model with no selection at all, and performing the same selection leakily on a classical learner reproduces them. The interventions themselves are therefore not what the reported improvement measures.

The practical conclusion is limited to this regime. For a dataset of this size, a classical ensemble on Morgan fingerprints is a sound default: it performed at least as well as every neural model we tested, and the only one that matched it needed pretraining on an external corpus and far more computation, while the ensemble itself keeps simpler inputs and fewer moving parts. Whether the balance changes with more

data is an open question that these results cannot address; larger datasets are generally expected to favor learned representations, but we did not measure a learning curve, and the point at which a graph network would overtake fixed fingerprints is likely to depend on the target and on how well the network is tuned.

4.2 Dimensionality, Conformational Overhead, and Model Efficiency

Richer molecular representations are only worth their cost if they add predictive information. Three-dimensional descriptors are the most expensive to obtain, since they require conformer embedding (ETKDGv3) and force-field minimization (MMFF) for every molecule. Our comparison separates what each block of features contributes (Section 3.2). The 2D physicochemical descriptors alone were clearly the weakest representation ($R^2 = 0.728$, against 0.769 for Morgan fingerprints; $\Delta R^2 = 0.041$, Holm-adjusted $p = 0.0081$). Adding fingerprints to them recovered the difference ($R^2 = 0.774$), and adding the ten 3D descriptors on top changed nothing detectable ($R^2 = 0.774$; $\Delta R^2 = 0.001$, $p = 0.8243$). Morgan fingerprints alone were within 0.005 of the 2D+3D+FP set in R^2 (Holm-adjusted $p = 1.0000$). The hybrid sets had a slightly lower MAE than Morgan alone ($\Delta = 0.006$), but this was not significant after correction ($p = 0.7689$). In this dataset the fingerprint block carries the predictive signal, and neither the descriptors nor the 3D block added a detectable amount to it. This conclusion concerns accuracy on this target; a ten-descriptor 3D summary could miss information that a richer 3D representation would capture.

The cost side is less clear-cut than usually assumed. For the 859 compounds, computing the features took 0.4 s for Morgan fingerprints and 6.9 s for the full 2D+3D+FP set on matrices of 2048 and 2275 features, respectively (wall-clock time with parallel workers on the workstation). The 3D set is the more expensive, but the absolute difference is a few seconds. The cost of a complete outer fold, with its inner architecture search, was instead dominated by model fitting: between 1182 and 2586 s on average across the four representations (1987 s for Morgan), without a clear relation to whether 3D descriptors were present (Figure 8). At this scale, then, conformer generation was not the bottleneck. We did not measure how the cost of 3D features grows for libraries of millions of compounds, where it could matter, and the deployed screening model does not use them: it takes only the 747 selected Morgan bits, which are computed from the 2D structure.

The parsimonious choice for this target is therefore the 2D fingerprint ensemble. This is a statement about information content and about the inputs the screening step needs, not a measured speed advantage; as shown in Section 3.3, the end-to-end benchmark of the classical protocol was not faster than the graph network.

4.3 Methodological Scope, Limitations, and Translational Potential

The reliability of any quantitative structure–activity relationship (QSAR) model is bounded by the chemical diversity of its training data. With 859 compounds, the applicability domain of the ensemble is restricted to the scaffolds and physicochemical

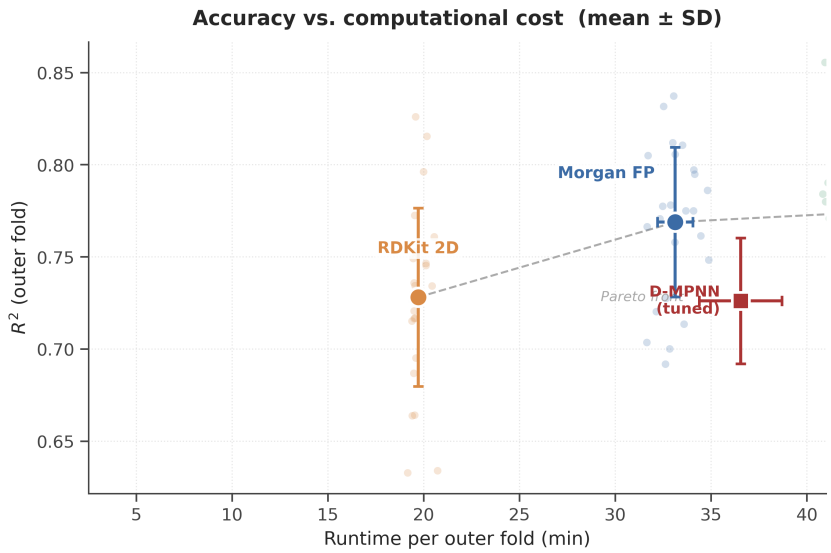


Fig. 8 Accuracy versus run time per outer fold (mean ± SD over the 25 folds; faint dots are individual folds). Only models that include a model-selection step are shown: the classical ensembles with their inner architecture search and the D-MPNN with a nested search over 8 configurations; the default D-MPNN, which has none, is reported in Table 4. Times are not normalized by thread count.

ranges of the curated TgDHFR set, and the FDA drugs are far from it: their median similarity to the five nearest training compounds is 0.17, against 0.70 for the training compounds among themselves, so most predicted potencies in the screening are extrapolations that the applicability domain cannot vouch for. The cross-validation estimate also needs a caveat: the main folds were random, and scaffold-grouped splits (Section 3.4) lowered R^2 for every model, by 0.175 for the Morgan ensemble (from 0.769 to 0.594) and 0.204 for the D-MPNN (from 0.725 to 0.521). The reported R^2 therefore describes interpolation among related compounds, and the expected accuracy on the structurally distant drugs of the FDA library is closer to the scaffold-split values. We did not evaluate temporal splits. The predicted potencies of those drugs have no experimental counterpart in this study.

Activity cliffs are a specific weakness of 2D circular fingerprints in this setting: very similar molecules can differ strongly in activity, and a Morgan bit vector may map them to nearly the same representation. We did not measure how many cliffs the data set contains or how the model behaves on them. The docking step does not remove the problem either; it adds an independent estimate, but a rigid-receptor Vina score is itself unreliable for small differences between close analogs, as the discordance for pyrimethamine in Section 3.6 illustrates.

The inner architecture search is a further source of instability. Evaluating 41 stacking combinations per outer fold carries a risk of selection bias. The nested design keeps this search away from the outer test folds, so the estimate is not inflated by it, but the chosen architecture varied across folds, and the Morgan space was the most settled of the four: its most frequent combination, RF+XGB+SVM, was selected in

76.0% of the 25 folds, against 32.0% for the least settled space (Figure S2 of the Supplementary Material). With a data set of this size the inner loop does not converge on a single architecture, and the reported performance should be read as that of the selection procedure and not of one fixed ensemble.

The applicability domain shows how little of this library the model can speak to: only 14 of the 3056 scored compounds reach the similarity threshold, so for the rest a predicted potency is an extrapolation whatever its value. That is the main reason to read this stage as hypothesis generation. It also bears on the ranking itself: the antifolates that lead it inhibit the human enzyme as well, and selectivity against human DHFR was not assessed here, so a high predicted potency is not by itself a reason to prefer a compound.

The most important translational limitation is selectivity. Pralatrexate, methotrexate and trimetrexate are anticancer antifolates that inhibit human dihydrofolate reductase, and are cytotoxic to host cells. Our pipeline predicts potency against the parasite enzyme and ranks docking energies against it, but it does not assess selectivity over the human enzyme, so a high rank should not be mistaken for a therapeutic candidate. The consensus hits are best read as computational hypotheses: compounds whose predicted activity is supported by two independent methods and that merit selectivity docking against human DHFR and experimental testing. The same pipeline could be applied to other targets with small data sets, but our results do not show that it transfers; that would need to be tested on those targets.

4.4 Conclusions and Future Directions

This study compared classical machine learning ensembles with a graph neural network on a small TgDHFR data set ($N = 859$), under a leakage-free nested cross-validation protocol, and applied the best classical model to a library of FDA-approved drugs with an orthogonal docking check. Three conclusions follow.

First, on this data set a classical stacking ensemble on 2D Morgan fingerprints performed at least as well as the alternatives we tested. It reached $R^2 = 0.769 \pm 0.041$, against 0.725 ± 0.043 for a default D-MPNN and 0.726 ± 0.034 for one tuned in a nested search; the difference ($\Delta R^2 = -0.043$ for the tuned network) was small but survived multiple-comparison correction, and equivalence with the Morgan ensemble could not be established in either metric, so the network as usually trained is behind the fingerprint-based ensembles rather than interchangeable with them. The gap is one of training budget and not of architecture: averaging five networks narrows it, and a D-MPNN initialized from a pretrained model reaches 0.761 and is the only model equivalent to the Morgan ensemble in both metrics, at about eight times the computational cost and without exceeding it. Adding 3D descriptors to the fingerprint-based sets changed nothing detectable. Feature selection and Gaussian-noise augmentation, which the original study credited with large gains, did not improve a classical learner under the same hold-out protocol. Those gains are accounted for by information leaks: removing them one at a time from the published code takes its R^2 from 0.853 to 0.740, and the same selection applied leakily to a classical learner reproduces them (0.836 against 0.748). These findings are limited to a single target and a comparison in which the network search was modest; they do not show that graph networks cannot match

fixed fingerprints. The classical protocol was also not cheaper to run: the benchmark took longer than the network's, and its advantage lies in accuracy that is not lower, simpler inputs, and a deployed model that needs only a subset of fingerprint bits.

Second, nested cross-validation and a careful curation of the data (removal of duplicates and of two extreme potency values, with a sensitivity analysis of the latter) were necessary to obtain estimates that can be compared across models. Our random-split estimates are also optimistic: with scaffold-grouped splits every model lost between 0.161 and 0.236 units of R^2 (to 0.594 for the Morgan ensemble and 0.521 for the D-MPNN), and the data could not resolve the differences between them. The loss was smallest for the pretrained network, so what representation learning offers on this data set is robustness to a change of chemotype rather than accuracy within one.

Third, combining ML prediction with docking produced a short, reproducible list of hypotheses and not validated leads. Only 14 of the 3056 scored compounds fell inside the model's applicability domain, the agreement between predicted potency and docking energy was weak ($\rho = -0.313$, $p = 0.2218$), and the compounds leading the ranking by potency are antifolates that also inhibit the human enzyme, whose selectivity was not assessed.

Future work should therefore (i) dock the consensus hits against human DHFR and test them in enzymatic assays against TgDHFR, (ii) evaluate temporal splits, (iii) compare against larger searches and pretrained graph models, and (iv) apply the pipeline to other targets before claiming that it generalizes.

Data and Code Availability

All source code, Python scripts for dataset processing, nested cross-validation, model training, and docking validation protocols, as well as the benchmark datasets, are publicly available in the GitHub repository at the following link

Declaration of Generative AI's use

During the preparation of this work, the author(s) utilized generative artificial intelligence technologies (Large Language Models) solely to assist with language polishing, structural formatting, and improving the overall readability of the manuscript. After using these tools, the author(s) thoroughly reviewed, edited, and validated all generated text. The author(s) take full and sole responsibility for the final content, scientific integrity, and accuracy of this publication. No AI technologies were used to generate data, perform statistical analyses, or draw scientific conclusions.

Acknowledgments

This work was supported in part by ANID FONDECYT Regular No. 1241977, FONDECYT Iniciación No. 11261195, and ANID CPS-RTC CIA250016 in Chile.

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